# Sandith Ganhewage

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## Education

**Duke University** — Durham, NC
*Bachelor's in Computer Science and Electrical & Computer Engineering* — Expected Grad: May 2028
- GPA: 3.98/4.00, Dean's List with Distinction (F24, S25), J. Welch Hariss Scholar

## Experience

**Amazon Web Services (AWS)** — May 2026 – Aug. 2026
*Software Development Engineer Intern* — Raleigh, NC
- Designing and implementing canary-style analytic pipeline stage for automated code promotion from live performance metrics

**o11** — Sep. 2025 – Mar. 2026
*Founding Engineer* — San Francisco, CA
- Built full-stack AI platform for bankers, enabling ingestion, embedding, and retrieval of proprietary financial documents
- Architected production RAG pipeline, implementing intelligent document chunking and embedding for scalable retrieval
- Developed Office 365 tools powered by RAG pipelines for document generation using firm-specific financial context
- Received backing from Y Combinator (W26), LeapYear, and KPMG, supporting enterprise pilots and product launch

**narb Inc.** — Aug. 2025 – Sep. 2025
*Software Engineering Intern* — Remote
- Designed user application for multi-modal LLM technology, utilizing Convex for a 30% increase in development speed
- Migrated existing applications to a new cross-platform framework prioritizing rapid development and implementation

**Qorvo Inc.** — May 2025 – Aug. 2025
*Test Engineering Intern* — Greensboro, NC
- Designed an embedded systems solution to chip sorting with the Synax S9 Handler, reducing configuration time by 25%
- Deployed company-wide solution for 50+ employees through an intranet-hosted web interface with React and Node.js
- Implemented low-level Arduino GPIB interface, alongside a Raspberry Pi connecting to universal web UI via bluetooth

**Duke Center for Computational and Evolutionary Intelligence** — Jan. 2025 – May 2025
*Undergraduate Researcher* — Durham, NC
- Tested and researched machine learning model quantization and distillation techniques to reduce model size by up to 30% and inference latency by up to 10% while maintaining model accuracy using PyTorch libraries
- Researched applications in 5 established bioinformatics algorithms for hardware acceleration by tri-state CAMs

## Projects

**Duke Automatic Code Grading Interface (ADTG)** | *Spring Boot, Thymeleaf, PostgreSQL*
- Developed automatic delivery, testing, and grading system alongside Duke Professor for 100+ student graduate classes
- Implemented grading infrastructure for instructors/TAs, building reliable backend logic for secure database transactions
- Diagnosed and resolved UI and database issues across the application, improving system reliability and user experience

**Low-Latency Matching Engine** | *C++, CMake, POSIX Sockets, Python, gtest*
- Built single-symbol limit order book in C++; profiling-driven optimizations, boosting throughput by 24% after local testing
- Replaced per-order heap allocation with a custom slab pool and lock-free producer-consumer ring with cache-aligned atomics, cutting worst-case latency by 41%
- Built TCP gateway with a fixed 40-byte binary message format shared between the network layer and internal queue, removing parsing overhead on the critical path
- Validated correctness with 20+ unit tests including a 5000-iteration randomized property test checking book state

**MeetUp** | *React, React Native, Tailwind CSS, Convex, Clerk, Google/Outlook OAuth*
- Built cross-platform scheduling app with Google/Outlook calendar integration, simplifying event coordination
- Implemented dynamic group scheduling and unified profiles to accelerate onboarding and collaboration
- Secured user authentication with Clerk and designed privacy controls for configurable data sharing across groups

## Extracurriculars

Duke Law School – *Instructional Technologies Consultant*, Durham, NC — Aug. 2025 - Dec. 2025
narbhacks – *Hackathon*, Remote — Jul. 2025
Duke Catalyst – *Pre-professional Tech Society*, Durham, NC — Jan. 2025 – Present
Duke Applied Machine Learning – *Hardware Division*, Durham, NC — Sep. 2024 – Dec. 2024
Joint School of Nanoscience and Nanoengineering – *Research Intern*, Greensboro, NC — Jul. 2023 – Jan. 2024

## Technical Skills

**Languages**: TypeScript, Python, Java, SQL, C/C++
**Frameworks & Libraries**: React, Next.js, Node.js, Spring Boot, PyTorch, shadcn/ui
**Databases & Cloud**: PostgreSQL, MongoDB, AWS, Azure, Docker
**Systems & Infrastructure**: CI/CD, Redis, BullMQ, RAG pipelines